QFT Notes

1. Attempts on Relativistic QM

In the case of **special relativity**, physics looks the same on every inertial frames. The spacetime coordinate (contravariant) is introduced as:

$$x^\mu = (x^0, x^1, x^2, x^3) = (ct, \boldsymbol{x}).$$

Minkowski metric, also known as flat spacetime tensor metric, is defined as:

$$g_{\mu\nu} = g^{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

So, the covariant form of the spacetime coordinate is just $x_\mu = g_{\mu\nu}x^\nu$. Another important property of the tensor metric is:

$$g^{\mu\nu}g_{\nu\rho} = \delta^\mu_\nu$$

which is just the delta Kronecker function.

Lorentz transformation is defined by these sets of equations.

$$\begin{aligned} c\bar{t} &= \gamma ct - \gamma\beta x, \\ \bar{x} &= -\gamma\beta ct + \gamma x, \\ \bar{y} &= y, \\ \bar{z} &= z. \end{aligned}$$

Then we can define Lorentz matrix as:

$$\Lambda^\mu_\nu = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Now, one can relate a transformation for the spacetime coordinate can be written as:

$$\bar{x}^\mu = \Lambda^\mu_\nu x^\nu + a^\mu,$$

where a^μ is just a constant translation vector. Lorentz transformation matrix must obey this equation:

$$g_{\mu\nu}\Lambda^\mu_\rho\Lambda^\nu_\sigma = g_{\rho\sigma},$$

which is straightforward to prove by contracting the indices μ and ν, we are then left with $\left(-\Lambda^0_\rho\Lambda^0_\sigma + \Lambda^1_\rho\Lambda^1_\sigma + \Lambda^2_\rho\Lambda^2_\sigma + \Lambda^3_\rho\Lambda^3_\sigma\right)$ which we can set as an unknown 4×4 matrix. If we solve for each matrix element, we will get that the matrix is equal to the tensor metric.

The equation ensures that the "length" between two points are the same on either frames.

$$\begin{aligned}(\bar{x}' - \bar{x})^2 &= g_{\mu\nu}(\bar{x}' - \bar{x})^\mu(\bar{x}' - \bar{x})^\nu \\ &= g_{\mu\nu}\Lambda^\mu_\rho\Lambda^\nu_\sigma(x' - x)^\rho(x' - x)^\sigma \\ &= g_{\rho\sigma}(x' - x)^\rho(x' - x)^\sigma \\ &= (x' - x)^2.\end{aligned}$$

In quantum mechanics, the time evolution of the state of a system is governed by one equation, the Schrodinger equation.

$$i\hbar\frac{\partial}{\partial t}|\psi,\, t\rangle = H|\psi,\, t\rangle.$$

Obvious attempt to consider the relativistic version is by setting the hamiltonian using the energy-momentum relation.

$$\begin{aligned}E^2 &= (pc)^2 + (mc^2)^2, \\ H &= +\sqrt{p^2c^2 + m^2c^4}.\end{aligned}$$

We can expand the hamiltonian using binomial expansion.

$$\begin{aligned}H &= +mc^2\sqrt{1 + \frac{p^2}{m^2c^2}} \\ &= +mc^2\left(1 + \frac{1}{2}\frac{p^2}{m^2c^2} - ...\right) \\ &= mc^2 + \frac{p^2}{2m} - ... \ .\end{aligned}$$

If we use this hamiltonian on the Schrodinger equation,

$$i\hbar\frac{\partial}{\partial t}\psi(\boldsymbol{x}, t) = +\sqrt{-\hbar^2c^2\nabla^2 + m^2c^4}\psi(\boldsymbol{x}, t),$$

square both side's operators then we get the **Klein-Gordon** equation,

$$-\hbar^2\frac{\partial^2}{\partial t^2}\psi(\boldsymbol{x}, t) = (-\hbar^2c^2\nabla^2 + m^2c^4)\psi(\boldsymbol{x}, t).$$

Without squaring both operators, there would be two problems:

1. Asymmetry of time and space. Time and space operators (in this case the differential) are separated, whereas in relativistic theory, time and space are one spacetime.
2. Non-locality in space. We already expanded hamiltonian using binomial expansion. We can see that the higher the order, there would be infinite number of differential of space (momentum operator).

To simplify this equation and bring the spacetime coordinate together, let us define a new differential operator.

$$\partial_\mu \equiv \frac{\partial}{\partial x^\mu} = \left(\frac{1}{c}\frac{\partial}{\partial t}, \nabla\right) ,$$
$$\partial^\mu \equiv \frac{\partial}{\partial x_\mu} = \left(-\frac{1}{c}\frac{\partial}{\partial t}, \nabla\right) .$$

The operator satisfies $\partial^\mu x^\nu = g^{\mu\nu}$ and relates with Lorentz transformation as $\bar{\partial}^\mu = \Lambda^\mu_\nu \partial^\nu$.

We can rewrite the Klein-Gordon as

$$\left(\partial^\mu \partial_\mu - m^2c^2/\hbar^2\right)\psi(x) = 0.$$

Suppose that we are looking at the equation in different inertial frame $\bar{x}$.

$$\begin{aligned}
\left(\bar{\partial}^\mu \bar{\partial}_\mu - m^2c^2/\hbar^2\right)\bar{\psi}(\bar{x}) &= 0 \\
\left(g_{\mu\nu}\bar{\partial}^\mu \bar{\partial}^\nu - m^2c^2/\hbar^2\right)\bar{\psi}(\bar{x}) &= 0 \\
\left(g_{\mu\nu}\Lambda^\mu_\rho \Lambda^\nu_\sigma \partial^\rho \partial^\sigma - m^2c^2/\hbar^2\right)\bar{\psi}(\bar{x}) &= 0 \\
\left(g_{\rho\sigma}\partial^\rho \partial^\sigma - m^2c^2/\hbar^2\right)\bar{\psi}(\bar{x}) &= 0 \\
\left(\partial^\mu \partial_\mu - m^2c^2/\hbar^2\right)\bar{\psi}(\bar{x}) &= 0 \ .
\end{aligned}$$

So, in both inertial frames, the wavefunctions satisfy the same Klein-Gordon equation.

Now that the Klein-Gordon equation satisfy conditions on the relativistic side, it actually violates the original Schrodinger equation. Notice how the time differential operator is in the first order but it is in the second order in Klein-Gordon equation. One of the prominent problem is that, the probability or norm of a state would then have some dependency on time (unconserved probability).

Dirac's attempt at solving this inconsistency starts with writing Schrodinger equation in the form

$$i\hbar\frac{\partial}{\partial t}\psi_a(x) = \left(-i\hbar c(\alpha^j)_{ab}\partial_j + mc^2\beta_{ab}\right)\psi_b(x)$$

where α^j and β are called the Dirac matrices in spin space and a and b are either 1 or 2. Dirac matrices are defined to obey these conditions

$$\left\{\alpha^j, \alpha^k\right\}_{ab} = 2\delta^{jk}\delta_{ab} \ , \quad \left\{\alpha^j, \beta\right\}_{ab} = 0 \ , \quad \left(\beta^2\right)_{ab} = \delta_{ab} \ .$$

Let us prove that these matrices must be even dimensional. First, we must find the eigenvalue of β using the third condition.

$$\begin{aligned}
\beta|b\rangle &= b|b\rangle \\
\beta^2|b\rangle &= b^2|b\rangle \ .
\end{aligned}$$

We know from the condition that β^2 in matrix form is just the identity matrix, so

$$b^2 = 1 \quad \rightarrow \quad b = \pm 1 \ .$$

Now all that is left is to make sure that $\text{Trace}(\beta) = 0$. To do that, let us consider $\{\alpha^j, \beta\}$.

$$\begin{aligned} \{\alpha_j, \beta\} &= 0 \\ \alpha_j\beta + \beta\alpha_j &= 0 \\ \alpha_j\alpha_j\beta + \alpha_j\beta\alpha_j &= 0 \\ \text{Trace}(\alpha_j\alpha_j\beta) + \text{Trace}(\alpha_j\beta\alpha_j) &= 0 \\ \text{Trace}(\alpha_j^2\beta) &= 0 \ . \end{aligned}$$

Now, consider the first condition. let $j = k$ and $a = b$ (diagonal matrix condition).

$$\begin{aligned} 2\alpha_j^2 &= 2 \\ \alpha_j^2 &= 1 \\ \alpha_j^2\beta &= \beta \\ \text{Trace}(\alpha_j^2\beta) &= \text{Trace}(\beta) = 0 \end{aligned}$$

For trace of α matrices, consider the second condition again. This time, instead of multiplying another alpha matrix, we multiply it with the inverse of β. The cyclical property of trace still applies. Then we can get that $\text{Trace}(\alpha_j) = 0$. Now, we have shown that the dirac matrices are even dimensional. As for the minimum dimension, Dirac matrices cannot be a 2×2 matrices. By working the anticommutator with the smallest matrices that satisfy $\{\alpha^j, \alpha^k\} = 2\delta^{jk}$, the Pauli matrices, we get that the only matrix capable of anticommuting with Pauli matrices is a zero matrix, or in another words, no "fourth" Pauli matrix exists. Therefore, their size must be 4×4.

From the traces of Dirac matrices, we can conclude that the trace of the hamiltonian ($H = c\alpha \cdot \boldsymbol{p} + mc^2\beta$) is also zero. So, the four eigenvalues are $+E(\boldsymbol{p})$, $+E(\boldsymbol{p})$, $-E(\boldsymbol{p})$, and $-E(\boldsymbol{p})$. Dirac's proposed solutions for the negative energy level: that by Pauli exclusion principles, those negative energy states are already occupied by electrons forming a sea of electrons.

To make a more generalized quantum theory in relativistic limits, we have to start over from the ground, but this time we put time and space in an equal footing. To accomplish that, one of the method is to just demote position (space) from operator to just parameter (like time in non-relativistic quantum mechanics).

Before getting into relativistic quantum mechancis we would like to "transform" ordinary quantum mechanics (non-relativistic) to field theory. We begin by introducing **quantum field** operator $a(\boldsymbol{x})$ and its conjugate $a^\dagger(\boldsymbol{x})$. For bosonic fields, the operators are subject to such rules:

$$[a(\boldsymbol{x}), a(\boldsymbol{x}')] = 0 \ , \quad \left[a^\dagger(\boldsymbol{x}), a^\dagger(\boldsymbol{x}')\right] = 0 \ , \quad \left[a(\boldsymbol{x}), a^\dagger(\boldsymbol{x}')\right] = \delta(\boldsymbol{x} - \boldsymbol{x}') \ .$$

Then, we can introduce the quantum field hamiltonian and state as

$$H = \int \mathrm{d}^3x a^\dagger(\boldsymbol{x})\left(-\frac{\hbar}{2m}\nabla^2+U(\boldsymbol{x})\right)a(\boldsymbol{x}) + \frac{1}{2}\int \mathrm{d}^3x\,\mathrm{d}^3y V(\boldsymbol{x}-\boldsymbol{y})a^\dagger(\boldsymbol{x})a^\dagger(\boldsymbol{y})a(\boldsymbol{y})a(\boldsymbol{x}) \ ,$$

$$|\psi,\, t\rangle = \int \mathrm{d}^3x_1 ... \,\mathrm{d}^3x_n \psi(\boldsymbol{x}_1, ..., \boldsymbol{x}_n; t)a^\dagger(\boldsymbol{x}_1)...a^\dagger(\boldsymbol{x}_n)|0\rangle \ .$$

We can apply the hamiltonian directly to the state ket. If we look at the form of hamiltonian, there are two elements of summation. We can call these as H_1 and H_2. We then can apply them separately to the state ket for the sake of simplicity. Before that, let us consider this equation.

$$\begin{aligned}
&a(\boldsymbol{x})a^\dagger(\boldsymbol{x}_1)a^\dagger(\boldsymbol{x}_2)...a^\dagger(\boldsymbol{x}_n)|0\rangle \\
&\quad = \big(a^\dagger(\boldsymbol{x}_1)a(\boldsymbol{x}) + \delta(\boldsymbol{x}-\boldsymbol{x}_1)\big)a^\dagger(\boldsymbol{x}_2)...a^\dagger(\boldsymbol{x}_n)|0\rangle \\
&\quad = a^\dagger(\boldsymbol{x}_1)a(\boldsymbol{x})a^\dagger(\boldsymbol{x}_2)...a^\dagger(\boldsymbol{x}_n)|0\rangle + \delta(\boldsymbol{x}-\boldsymbol{x}_1)a^\dagger(\boldsymbol{x}_2)...a^\dagger(\boldsymbol{x}_n)|0\rangle \\
&\quad \vdots \\
&\quad = a^\dagger(\boldsymbol{x}_1)...a^\dagger(\boldsymbol{x}_n)a(\boldsymbol{x})|0\rangle + \sum_{j=1}^{n}\delta\big(x-x_j\big)\prod_{k\neq j}^{n}a^\dagger(\boldsymbol{x}_n)|0\rangle \\
&\quad = \sum_{j=1}^{n}\delta\big(\boldsymbol{x}-\boldsymbol{x}_j\big)\prod_{k\neq j}^{n}a^\dagger(\boldsymbol{x}_k)|0\rangle \ .
\end{aligned}$$

Now, we can consider the H_1 operator.

$$\begin{aligned}
&H_1|\psi,\, t\rangle \\
&\quad = \int \mathrm{d}^3x\,\mathrm{d}^3x_1 ... \,\mathrm{d}^3x_n a^\dagger(\boldsymbol{x})\left(-\frac{\hbar}{2m}\nabla^2+U(\boldsymbol{x})\right)\psi(\boldsymbol{x}_1, ..., \boldsymbol{x}_n; t)a(\boldsymbol{x})a^\dagger(\boldsymbol{x}_1)...a^\dagger(\boldsymbol{x}_n)|0\rangle \\
&\quad = \int \mathrm{d}^3x\,\mathrm{d}^3x_1 ... \,\mathrm{d}^3x_n a^\dagger(\boldsymbol{x})K(\boldsymbol{x})\psi\left(\sum_{j=1}^{n}\delta\big(\boldsymbol{x}-\boldsymbol{x}_j\big)\prod_{k\neq j}^{n}a^\dagger(\boldsymbol{x}_k)|0\rangle\right) \\
&\quad = \sum_{j=1}^{n}\int \mathrm{d}x_1 ... \,\mathrm{d}x_n a^\dagger(\boldsymbol{x}_1)...a^\dagger(\boldsymbol{x}_n)\big(K_j\big(\boldsymbol{x}_j\big)\psi\big)|0\rangle \\
&\quad = \int \mathrm{d}x_1 ... \,\mathrm{d}x_n a^\dagger(\boldsymbol{x}_1)...a^\dagger(\boldsymbol{x}_n)\sum_{j=1}^{n}\left(-\frac{\hbar}{2m}\nabla_j^2+U\big(\boldsymbol{x}_j\big)\right)\psi|0\rangle
\end{aligned}$$

Now, let us consider this equation first before moving on to H_2.

$$\begin{aligned}
&a(\boldsymbol{y})a(\boldsymbol{x})a^\dagger(\boldsymbol{x}_1)a^\dagger(\boldsymbol{x}_2)...a^\dagger(\boldsymbol{x}_n)|0\rangle \\
&\qquad = \sum_{j=1}^{n}\delta\big(\boldsymbol{x}-\boldsymbol{x}_j\big)a(\boldsymbol{y})\prod_{k\neq j}^{n}a^\dagger(\boldsymbol{x}_n)|0\rangle \\
&\qquad = \sum_{j=1}^{n}\delta\big(\boldsymbol{x}-\boldsymbol{x}_j\big)\sum_{k\neq j}^{n}\delta(\boldsymbol{y}-\boldsymbol{x}_k)\prod_{l\neq k,j}^{n}a^\dagger(\boldsymbol{x}_l)|0\rangle
\end{aligned}$$

Let us consider H_2 operator.

$$
\begin{aligned}
&H_2|\psi,\, t\rangle \\
&= \int \mathrm{d}^3x\, \mathrm{d}^3y\, \mathrm{d}^3x_1 ... \, \mathrm{d}^3x_n V(\boldsymbol{x}-\boldsymbol{y})\psi a^\dagger(\boldsymbol{x})a^\dagger(\boldsymbol{y})a(\boldsymbol{y})a(\boldsymbol{x})a^\dagger(\boldsymbol{x}_1)a^\dagger(\boldsymbol{x}_2)...a^\dagger(\boldsymbol{x}_n)|0\rangle \\
&= \sum_{j=1}^{n}\sum_{k\neq j}^{n} \int \mathrm{d}^3x\, \mathrm{d}^3y\, \mathrm{d}^3x_1 ... \, \mathrm{d}^3x_n V(\boldsymbol{x}-\boldsymbol{y})\psi\delta\big(\boldsymbol{x}-\boldsymbol{x}_j\big)\delta(\boldsymbol{y}-\boldsymbol{x}_k)a^\dagger(\boldsymbol{x})a^\dagger(\boldsymbol{y}) \prod_{l\neq k,j}^{n} a^\dagger(\boldsymbol{x}_l)|0\rangle \\
&= \sum_{j=1}^{n}\sum_{k\neq j}^{n} \int \mathrm{d}^3x_1 ... \, \mathrm{d}^3x_n V\big(\boldsymbol{x}_j-\boldsymbol{x}_k\big)\psi a^\dagger\big(\boldsymbol{x}_j\big)a^\dagger(\boldsymbol{x}_k) \prod_{l\neq k,j}^{n} a^\dagger(\boldsymbol{x}_l)|0\rangle \\
&= \int \mathrm{d}^3x_1 ... \, \mathrm{d}^3x_n a^\dagger(\boldsymbol{x}_1)...a^\dagger(\boldsymbol{x}_n)\left(\sum_{j=1}^{n}\sum_{k\neq j}^{n} V\big(\boldsymbol{x}_j-\boldsymbol{x}_k\big)\right)\psi|0\rangle \ .
\end{aligned}
$$

Putting all back together,

$$
\begin{aligned}
H|\psi,\, t\rangle &= H_1|\psi,\, t\rangle + \frac{1}{2}H_2|\psi,\, t\rangle \\
&= \int \mathrm{d}^3x_1 ... \, \mathrm{d}^3x_n a^\dagger(\boldsymbol{x}_1)...a^\dagger(\boldsymbol{x}_n)\left(\sum_{j=1}^{n}\left(-\frac{\hbar}{2m}\nabla_j^2+U\big(\boldsymbol{x}_j\big)\right) + \frac{1}{2}\sum_{j=1}^{n}\sum_{k\neq j}^{n} V\big(\boldsymbol{x}_j-\boldsymbol{x}_k\big)\right)\psi|0\rangle \\
&= \int \mathrm{d}^3x_1 ... \, \mathrm{d}^3x_n a^\dagger(\boldsymbol{x}_1)...a^\dagger(\boldsymbol{x}_n)\left(\sum_{j=1}^{n}\left(-\frac{\hbar}{2m}\nabla_j^2+U\big(\boldsymbol{x}_j\big)\right) + \sum_{j=1}^{n}\sum_{k=1}^{j-1} V\big(\boldsymbol{x}_j-\boldsymbol{x}_k\big)\right)\psi|0\rangle \ .
\end{aligned}
$$

Applying time operator to the state,

$$
i\hbar\frac{\partial}{\partial t}|\psi,\, t\rangle = \int \mathrm{d}^3x_1 ... \, \mathrm{d}^3x_n a^\dagger(\boldsymbol{x}_1)...a^\dagger(\boldsymbol{x}_n)\left(i\hbar\frac{\partial}{\partial t}\right)\psi|0\rangle \ .
$$

If we demand that the hamiltonial satisfy the Schrodinger equation, then

$$
\begin{aligned}
i\hbar\frac{\partial}{\partial t}|\psi,\, t\rangle &= H|\psi,\, t\rangle \\
i\hbar\frac{\partial\psi}{\partial t} &= \left[\sum_{j=1}^{n}\left(-\frac{\hbar}{2m}\nabla_j^2+U\big(\boldsymbol{x}_j\big)\right) + \sum_{j=1}^{n}\sum_{k=1}^{j-1} V\big(\boldsymbol{x}_j-\boldsymbol{x}_k\big)\right]\psi \ .
\end{aligned}
$$

Look at the integration from the state ket. Having wave function with space dependency and that creation operator commutes with each other, we can see that only symmetric ψ would survive the integration.

$$
\psi\big(...\boldsymbol{x}_i...\boldsymbol{x}_j...;t\big) = \psi\big(...\boldsymbol{x}_j...\boldsymbol{x}_i...;t\big) \ .
$$

Fermionic field, on the other hand, obeys these anticommutator relation:

$$
\{a(\boldsymbol{x}), a(\boldsymbol{x}')\} = 0 \ , \quad \big\{a^\dagger(\boldsymbol{x}), a^\dagger(\boldsymbol{x}')\big\} = 0 \ , \quad \big\{a(\boldsymbol{x}), a^\dagger(\boldsymbol{x}')\big\} = 0 \ .
$$

In contrast with the previous bosonic field, the fermionic wave equation's part that would survive the integration in state ket equation is the antisymmetric part.

$$
\psi\big(...\boldsymbol{x}_i...\boldsymbol{x}_j...;t\big) = -\psi\big(...\boldsymbol{x}_j...\boldsymbol{x}_i...;t\big) \ .
$$

There is another operator that counts the number of particles, that is

$$N = \int \mathrm{d}^3 x a^\dagger(\boldsymbol{x}) a(\boldsymbol{x}) \ .$$

We can see how this commutes with previous hamiltonian for fields.

$$[N, H] = NH - HN \ ,$$

$$\begin{aligned}
NH &= \int \mathrm{d}^3 x' \, \mathrm{d}^3 x a^\dagger(\boldsymbol{x}') a(\boldsymbol{x}') a^\dagger(\boldsymbol{x}) K(\boldsymbol{x}) a(\boldsymbol{x}) \\
&\quad + \frac{1}{2} \int \mathrm{d}^3 x' \, \mathrm{d}^3 x \, \mathrm{d}^3 y V(\boldsymbol{x} - \boldsymbol{y}) a^\dagger(\boldsymbol{x}') a(\boldsymbol{x}') a^\dagger(\boldsymbol{x}) a^\dagger(\boldsymbol{y}) a(\boldsymbol{y}) a(\boldsymbol{x}) \\
&= \int \mathrm{d}^3 x' \, \mathrm{d}^3 x a^\dagger(\boldsymbol{x}) a^\dagger(\boldsymbol{x}') K(\boldsymbol{x}) a(\boldsymbol{x}') a(\boldsymbol{x}) + \int \mathrm{d}^3 x a^\dagger(\boldsymbol{x}) K(\boldsymbol{x}) a(\boldsymbol{x}) \\
&\quad + \frac{1}{2} \int \mathrm{d}^3 x' \, \mathrm{d}^3 x \, \mathrm{d}^3 y V(\boldsymbol{x} - \boldsymbol{y}) a^\dagger(\boldsymbol{x}) a^\dagger(\boldsymbol{y}) a^\dagger(\boldsymbol{x}') a(\boldsymbol{x}') a(\boldsymbol{y}) a(\boldsymbol{x}) \\
&\quad + \int \mathrm{d}^3 x \, \mathrm{d}^3 y V(\boldsymbol{x} - \boldsymbol{y}) a^\dagger(\boldsymbol{x}) a^\dagger(\boldsymbol{y}) a(\boldsymbol{y}) a(\boldsymbol{x}) \ , \\
HN &= \int \mathrm{d}^3 x \, \mathrm{d}^3 x' a^\dagger(\boldsymbol{x}) K(\boldsymbol{x}) a(\boldsymbol{x}) a^\dagger(\boldsymbol{x}') a(\boldsymbol{x}') \\
&\quad + \frac{1}{2} \int \mathrm{d}^3 x \, \mathrm{d}^3 y \, \mathrm{d}^3 x' V(\boldsymbol{x} - \boldsymbol{y}) a^\dagger(\boldsymbol{x}) a^\dagger(\boldsymbol{y}) a(\boldsymbol{y}) a(\boldsymbol{x}) a^\dagger(\boldsymbol{x}') a(\boldsymbol{x}') \\
&= \int \mathrm{d}^3 x' \, \mathrm{d}^3 x a^\dagger(\boldsymbol{x}) a^\dagger(\boldsymbol{x}') K(\boldsymbol{x}) a(\boldsymbol{x}') a(\boldsymbol{x}) + \int \mathrm{d}^3 x a^\dagger(\boldsymbol{x}) K(\boldsymbol{x}) a(\boldsymbol{x}) \\
&\quad + \frac{1}{2} \int \mathrm{d}^3 x' \, \mathrm{d}^3 x \, \mathrm{d}^3 y V(\boldsymbol{x} - \boldsymbol{y}) a^\dagger(\boldsymbol{x}) a^\dagger(\boldsymbol{y}) a^\dagger(\boldsymbol{x}') a(\boldsymbol{x}') a(\boldsymbol{y}) a(\boldsymbol{x}) \\
&\quad + \int \mathrm{d}^3 x \, \mathrm{d}^3 y V(\boldsymbol{x} - \boldsymbol{y}) a^\dagger(\boldsymbol{x}) a^\dagger(\boldsymbol{y}) a(\boldsymbol{y}) a(\boldsymbol{x}) \ .
\end{aligned}$$

Thus,

$$[N, H] = 0 \ .$$

2. Lorentz Invariance

The set of all Lorentz transformation forms a *group*:

1. The product of any two Lorentz transformations is another Lorentz transformation,
2. The product is associative,
3. There is an identity transformation, $\Lambda^{\mu}{}_{\nu} = \delta^{\mu}{}_{\nu}$,
4. Every Lorentz transformation matrix has an inverse.

Let us look at the definition of Lorentz transformation in spacetime coordinate:

$$\bar{x}^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu} \ .$$

If we go back to matrix notation, to transform x to $\bar{x}$, we multiply the inverse of matrix Λ from the left side of the equation. So that in the end we would get

$$x^{\mu} = \left(\Lambda^{-1}\right)^{\mu}{}_{\nu} \bar{x}^{\nu} \ .$$

The equivalent of this would be applying $\left(\Lambda^{-1}\right)^{\rho}{}_{\mu}$ to the Lorentz transformation definition.

$$\left(\Lambda^{-1}\right)^{\rho}{}_{\mu} \bar{x}^{\mu} = \left(\Lambda^{-1}\right)^{\rho}{}_{\mu} \Lambda^{\mu}{}_{\nu} x^{\nu} \ .$$

So, in order to get the final expression, we must assert that

$$\left(\Lambda^{-1}\right)^{\rho}{}_{\mu} \Lambda^{\mu}{}_{\nu} = \delta^{\rho}{}_{\nu} \ .$$

Using that identity, we can go back to the identity between Lorentz transformaton and Minkowski metric to find how we can express the inverse of a Lorentz transformation in the form of a regular Lorentz matrix.

$$\begin{aligned} g_{\mu\nu} \Lambda^{\mu}{}_{\rho} \Lambda^{\nu}{}_{\sigma} &= g_{\rho\sigma} \\ \Lambda_{\nu\rho} \Lambda^{\nu}{}_{\sigma} &= g_{\rho\sigma} \\ \Lambda_{\nu}{}^{\mu} \Lambda^{\nu}{}_{\sigma} &= g^{\mu\rho} g_{\rho\sigma} \\ \Lambda_{\nu}{}^{\mu} \Lambda^{\nu}{}_{\sigma} = \delta^{\mu}{}_{\sigma} \rightarrow \Lambda_{\nu}{}^{\rho} \Lambda^{\nu}{}_{\sigma} &= \delta^{\rho}{}_{\sigma} \ . \end{aligned}$$

From the previous identity,

$$\left(\Lambda^{-1}\right)^{\rho}{}_{\nu} = \Lambda_{\nu}{}^{\rho} = g_{\alpha\nu} g^{\beta\rho} \Lambda^{\alpha}{}_{\beta} \quad \rightarrow \quad \left(\Lambda^{-1}\right)^{\mu}{}_{\nu} = \Lambda_{\nu}{}^{\mu} = g_{\nu\alpha} g^{\mu\beta} \Lambda^{\alpha}{}_{\beta}$$

We can also express

$$
\begin{aligned}
g_{\mu\nu}(\Lambda^{-1})^{\mu}{}_{\rho}(\Lambda^{-1})^{\nu}{}_{\sigma} &= g_{\rho\sigma} \\
g_{\mu\nu}\Lambda_{\rho}{}^{\mu}\Lambda_{\sigma}{}^{\nu} &= g_{\rho\sigma} \\
g_{\mu\nu}g_{\rho\alpha}g^{\mu\beta}\Lambda^{\alpha}{}_{\beta}g_{\sigma\gamma}g^{\nu\varepsilon}\Lambda^{\gamma}{}_{\varepsilon} &= g_{\rho\sigma} \\
g_{\rho\alpha}g_{\sigma\gamma}\left(g^{\beta\mu}g_{\mu\nu}\right)g^{\nu\varepsilon}\Lambda^{\alpha}{}_{\beta}\Lambda^{\gamma}{}_{\varepsilon} &= g_{\rho\sigma} \\
g_{\rho\alpha}g_{\sigma\gamma}\delta^{\beta}{}_{\nu}g^{\nu\varepsilon}\Lambda^{\alpha}{}_{\beta}\Lambda^{\gamma}{}_{\varepsilon} &= g_{\rho\sigma} \\
g_{\rho\alpha}g_{\sigma\gamma}g^{\beta\varepsilon}\Lambda^{\alpha}{}_{\beta}\Lambda^{\gamma}{}_{\varepsilon} &= g_{\rho\sigma} \ .
\end{aligned}
$$

Then, we must assert that

$$
g^{\beta\varepsilon}\Lambda^{\alpha}{}_{\beta}\Lambda^{\gamma}{}_{\varepsilon} = g^{\alpha\gamma} \quad \rightarrow \quad g^{\mu\nu}\Lambda^{\rho}{}_{\mu}\Lambda^{\sigma}{}_{\nu} = g^{\rho\sigma} \ ,
$$

which is just the contravariant version for the Lorentz metric identity.

An infinitesimal Lorentz transformation can be written as

$$
\Lambda^{\mu}{}_{\nu} = \delta^{\mu}{}_{\nu} + \delta\omega^{\mu}{}_{\nu} \ .
$$

Going back to the Lorentz metric identity, we will see the relation of $\delta\omega$ when the indices are swapped.

$$
\begin{aligned}
g_{\mu\nu}\left(\delta^{\mu}{}_{\rho} + \delta\omega^{\mu}{}_{\rho}\right)\left(\delta^{\nu}{}_{\sigma} + \delta\omega^{\nu}{}_{\sigma}\right) &= g_{\rho\sigma} \\
g_{\mu\nu}\left(\delta^{\mu}{}_{\rho}\delta^{\nu}{}_{\sigma} + \delta^{\mu}{}_{\rho}\delta\omega^{\nu}{}_{\sigma} + \delta\omega^{\mu}{}_{\rho}\delta^{\nu}{}_{\sigma} + \delta\omega^{\mu}{}_{\rho}\delta\omega^{\nu}{}_{\sigma}\right) &= g_{\rho\sigma} \\
g_{\rho\sigma} + \delta\omega_{\rho\sigma} + \delta\omega_{\sigma\rho} + g_{\mu\nu}\delta\omega^{\mu}{}_{\rho}\delta\omega^{\nu}{}_{\sigma} &= g_{\rho\sigma} \\
\delta\omega_{\rho\sigma} &= -\delta\omega_{\sigma\rho} \ .
\end{aligned}
$$

Running the indices, we can see that there are six independent infinitesimal Lorentz transformation in spacetime coordinate (triangular parts of the matrix) and those are divided into three translations and three rotations.

Lorentz transformation are **proper** when the determinant of the matrix is equal to $+1$ and **improper** when the determinant is equal to -1. There is also another subgroup called the **orthochronous** Lorentz transformation. It's when the element $\Lambda^{0}{}_{0} \geq +1$. A Lorentz transformation that can be obtained by compounding numbers of infinitesimal Lorentz transformation is called orthochronous and proper. In contrast of this, there are also nonorthochronous and improper Lorentz transformation such as

$$
\mathcal{P}^{\mu}{}_{\nu} = (\mathcal{P}^{-1})^{\mu}{}_{\nu} = \begin{pmatrix} +1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \qquad \mathcal{T}^{\mu}{}_{\nu} = (\mathcal{T}^{-1})^{\mu}{}_{\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & +1 & 0 & 0 \\ 0 & 0 & +1 & 0 \\ 0 & 0 & 0 & +1 \end{pmatrix}
$$

In quantum theory, symmetries are representated by unitary operators. For infinitesimal Lorentz transformation, we could write the unitary operator as

$$U(1+\delta\omega) = \mathbb{1} + \frac{i}{2\hbar}\delta\omega_{\mu\nu}M^{\mu\nu} \ ,$$

where M is the generator of Lorentz group with it satisfying $M^{\mu\nu} = -M^{\nu\mu}$. Unitary operators have the composition rule which state that $U(\Lambda'\Lambda) = U(\Lambda')U(\Lambda)$. Taking $\Lambda' = 1 + \delta\omega'$, we can expand $U(\Lambda)^{-1}U(\Lambda')U(\Lambda) = U(\Lambda^{-1}\Lambda'\Lambda)$ as

$$\begin{aligned}
&U(\Lambda)^{-1}U(\Lambda')U(\Lambda) \\
&= U(\Lambda)^{-1}U(1+\delta\omega')U(\Lambda) \\
&= U(\Lambda)^{-1}U(\Lambda) + \frac{i}{2\hbar}U(\Lambda)^{-1}\delta\omega'_{\ \mu\nu}M^{\mu\nu}U(\Lambda) \\
&= \mathbb{1} + \frac{i}{2\hbar}U(\Lambda)^{-1}\delta\omega'_{\ \mu\nu}M^{\mu\nu}U(\Lambda)
\end{aligned}$$

$$\begin{aligned}
&\Lambda^{-1}\Lambda'\Lambda \\
&= (\Lambda^{-1})^{\mu}_{\ \ \rho}(\Lambda')^{\rho}_{\ \ \sigma}\Lambda^{\sigma}_{\ \ \nu} \\
&= \Lambda_{\rho}^{\ \mu}(\Lambda')^{\rho}_{\ \ \sigma}\Lambda^{\sigma}_{\ \ \nu} \\
&= g_{\rho\alpha}\Lambda^{\alpha\mu}(\Lambda')^{\rho}_{\ \ \sigma}\Lambda^{\sigma}_{\ \ \nu} \\
&= (\Lambda^{-1})^{\mu}_{\ \ \rho}\delta^{\rho}_{\ \sigma}\Lambda^{\sigma}_{\ \ \nu} + g_{\rho\alpha}\Lambda^{\alpha\mu}(\delta\omega')^{\rho}_{\ \ \sigma}\Lambda^{\sigma}_{\ \ \nu} \\
&= \delta^{\mu}_{\ \ \nu} + \Lambda^{\alpha\mu}(\delta\omega')_{\alpha\sigma}\Lambda^{\sigma}_{\ \ \nu} \\
&= \delta^{\mu}_{\ \ \nu} + \delta\Omega^{\mu}_{\ \ \nu}
\end{aligned}$$

So,

$$\begin{aligned}
U(\Lambda)^{-1}U(\Lambda')U(\Lambda) &= U(\Lambda^{-1}\Lambda'\Lambda) \\
\mathbb{1} + \frac{i}{2\hbar}U(\Lambda)^{-1}\delta\omega'_{\ \mu\nu}M^{\mu\nu}U(\Lambda) &= U(1+\delta\Omega) \\
\mathbb{1} + \frac{i}{2\hbar}U(\Lambda)^{-1}\delta\omega'_{\ \mu\nu}M^{\mu\nu}U(\Lambda) &= \mathbb{1} + \frac{i}{2\hbar}\delta\Omega_{\mu\nu}M^{\mu\nu} \\
U(\Lambda)^{-1}\delta\omega'_{\ \mu\nu}M^{\mu\nu}U(\Lambda) &= \Lambda^{\mu}_{\ \ \alpha}\Lambda^{\nu}_{\ \ \beta}\delta\omega'_{\ \mu\nu}M^{\alpha\beta} \\
U(\Lambda)^{-1}M^{\mu\nu}U(\Lambda) &= \Lambda^{\mu}_{\ \ \alpha}\Lambda^{\nu}_{\ \ \beta}M^{\alpha\beta} \ .
\end{aligned}$$

If we set $\Lambda = 1 + \delta\omega$, we could get the commutation expression of Lorentz generators. Let us take a look at the LHS part of the last equation.

$$\begin{aligned}
U(\Lambda)^{-1}M^{\mu\nu}U(\Lambda) &= \left(\mathbb{1} - \frac{i}{2\hbar}\delta\omega_{\alpha\beta}M^{\alpha\beta}\right)M^{\mu\nu}\left(\mathbb{1} + \frac{i}{2\hbar}\delta\omega_{\rho\sigma}M^{\rho\sigma}\right) \\
&= M^{\mu\nu} + \frac{i}{2\hbar}\delta\omega_{\rho\sigma}\left[M^{\mu\nu}, M^{\rho\sigma}\right] + \frac{1}{4\hbar^2}\delta\omega_{\alpha\beta}\delta\omega_{\rho\sigma}M^{\alpha\beta}M^{\mu\nu}M^{\rho\sigma}
\end{aligned}$$

Now, let's look at the RHS part.

$$\begin{aligned}
\Lambda^{\mu}_{\ \ \rho}\Lambda^{\nu}_{\ \ \sigma}M^{\rho\sigma} &= \left(\delta^{\mu}_{\ \ \rho} + \delta\omega^{\mu}_{\ \ \rho}\right)(\delta^{\nu}_{\ \ \sigma} + \delta\omega^{\nu}_{\ \ \sigma})M^{\rho\sigma} \\
&= \left(\delta^{\mu}_{\ \ \rho}\delta^{\nu}_{\ \ \sigma} + \delta^{\mu}_{\ \ \rho}\delta\omega^{\nu}_{\ \ \sigma} + \delta\omega^{\mu}_{\ \ \rho}\delta^{\nu}_{\ \ \sigma} + \delta\omega^{\mu}_{\ \ \rho}\delta\omega^{\nu}_{\ \ \sigma}\right)M^{\rho\sigma} \\
&= M^{\mu\nu} + \delta\omega^{\nu}_{\ \ \beta}M^{\mu\beta} + \delta\omega^{\mu}_{\ \ \beta}M^{\beta\nu} + \delta\omega^{\mu}_{\ \ \rho}\delta\omega^{\nu}_{\ \ \sigma}M^{\rho\sigma} \\
&= M^{\mu\nu} + g^{\nu\alpha}\delta\omega_{\alpha\beta}M^{\mu\beta} + g^{\mu\alpha}\delta\omega_{\alpha\beta}M^{\beta\nu} - g^{\mu\sigma}g^{\nu\alpha}\delta\omega_{\alpha\beta}\delta\omega_{\rho\sigma}M^{\rho\beta}
\end{aligned}$$

If we equate both sides and the antisymmetric part of the $\delta\omega$ coefficients,

$$
\begin{aligned}
\frac{i}{2\hbar}\delta\omega_{\rho\sigma}\big[M^{\mu\nu},M^{\rho\sigma}\big] &= \delta\omega_{\rho\sigma}\big(g^{\nu\rho}M^{\mu\sigma}+g^{\mu\rho}M^{\sigma\nu}\big)\\
\frac{i}{2\hbar}\delta\omega_{\rho\sigma}\big[M^{\mu\nu},M^{\rho\sigma}\big] &= \frac{1}{2}\delta\omega_{\rho\sigma}\big(g^{\nu\rho}M^{\mu\sigma}+g^{\mu\rho}M^{\sigma\nu}\big)+\frac{1}{2}\delta\omega_{\rho\sigma}\big(g^{\nu\rho}M^{\mu\sigma}+g^{\mu\rho}M^{\sigma\nu}\big)\\
\frac{i}{2\hbar}\delta\omega_{\rho\sigma}\big[M^{\mu\nu},M^{\rho\sigma}\big] &= \frac{1}{2}\delta\omega_{\rho\sigma}\big(g^{\nu\rho}M^{\mu\sigma}+g^{\mu\rho}M^{\sigma\nu}\big)-\frac{1}{2}\delta\omega_{\rho\sigma}\big(g^{\nu\sigma}M^{\mu\rho}+g^{\mu\sigma}M^{\rho\nu}\big)\\
\big[M^{\mu\nu},M^{\rho\sigma}\big] &= i\hbar\big(g^{\nu\sigma}M^{\mu\rho}+g^{\mu\sigma}M^{\rho\nu}-g^{\nu\rho}M^{\mu\sigma}-g^{\mu\rho}M^{\sigma\nu}\big)\\
&= i\hbar\big(g^{\mu\rho}M^{\nu\sigma}-g^{\nu\rho}M^{\mu\sigma}+g^{\nu\sigma}M^{\mu\rho}-g^{\mu\sigma}M^{\nu\rho}\big)\ .
\end{aligned}
$$

These commutation relations specify the **Lie algebra** of Lorentz group. Let us take the angular momentum operator $J_i \equiv \frac{1}{2}\varepsilon_{ijk}M^{jk}$, so

$$
\begin{aligned}
\big[J_i,J_j\big] &= \frac{1}{4}\varepsilon_{iab}\varepsilon_{jcd}\big[M^{ab},M^{cd}\big]\\
&= \frac{i\hbar}{4}\varepsilon_{iab}\varepsilon_{jcd}\big(\delta^{ac}M^{bd}-\delta^{bc}M^{ad}+\delta^{bd}M^{ac}-\delta^{ad}M^{bc}\big)\\
&= \frac{i\hbar}{4}\big(\varepsilon_{icb}\varepsilon_{jcd}M^{bd}-\varepsilon_{iac}\varepsilon_{jcd}M^{ad}+\varepsilon_{iad}\varepsilon_{jcd}M^{ac}-\varepsilon_{idb}\varepsilon_{jcd}M^{bc}\big)\\
&= \frac{i\hbar}{4}\big(\varepsilon_{cbi}\varepsilon_{cdj}M^{bd}-\varepsilon_{cia}\varepsilon_{cdj}M^{ad}+\varepsilon_{dia}\varepsilon_{djc}M^{ac}-\varepsilon_{dbi}\varepsilon_{djc}M^{bc}\big)\ ,
\end{aligned}
$$

using the property of $\varepsilon_{ijk}\varepsilon_{imn}=\delta_{jm}\delta_{kn}-\delta_{jn}\delta_{km}$, we could get

$$
\begin{aligned}
\big[J_i,J_j\big] &= i\hbar\big(M^{bb}\delta_{ij}+M^{ij}\big)\\
&= i\hbar M^{ij}\ .
\end{aligned}
$$

Inversing the definition of angular momentum to get the generator M,

$$
\begin{aligned}
\varepsilon_{abi}J_i &= \frac{1}{2}\varepsilon_{abi}\varepsilon_{ijk}M^{jk}\\
&= \frac{1}{2}\varepsilon_{iab}\varepsilon_{ijk}M^{jk}\\
&= \frac{1}{2}\big(\delta_{aj}\delta_{bk}-\delta_{ak}\delta_{bj}\big)M^{jk}\\
&= \frac{1}{2}\big(M^{ab}-M^{ba}\big)\\
M^{ab} &= \varepsilon_{abk}J_k\ ,
\end{aligned}
$$

substituting it back to the commutation relation, we get that

$$
\big[J_i,J_j\big] = i\hbar\varepsilon_{ijk}J_k\ .
$$

Now, let us see how the angular momentum operator commutes with the boost operator $\boldsymbol{K}$ with its definition as $K\equiv M^{i0}$. From the commutation relation of M, we multiply with $\frac{1}{2}\varepsilon_{iab}$ and then we set $c,d=j,0$.

$$
\begin{aligned}
\big[J_i, K_j\big] &= \frac{1}{2}\varepsilon_{iab}\big[M^{ab}, M^{j0}\big] \\
&= \frac{i\hbar}{2}\varepsilon_{iab}\big(\delta^{aj}M^{b0} - \delta^{bj}M^{a0} + \delta^{b0}M^{aj} - \delta^{a0}M^{bj}\big) \\
&= \frac{i\hbar}{2}\big(\varepsilon_{ijb}K_b - \varepsilon_{iaj}K^a\big) \\
&= \frac{i\hbar}{2}\big(\varepsilon_{ijb}K_b + \varepsilon_{ija}K^a\big) \\
&= i\hbar\varepsilon_{ijk}K_k \ .
\end{aligned}
$$

Now, let's see how boost operator commutes with itself.

$$
\begin{aligned}
\big[K_i, K_j\big] &= \big[M^{i0}, M^{j0}\big] \\
&= i\hbar\big(\delta^{ij}M^{00} - g^{0j}M^{i0} + g^{00}M^{ij} - g^{i0}M^{0j}\big) \\
&= -i\hbar M^{ij} \\
&= -i\hbar\varepsilon_{ijk}J_k \ .
\end{aligned}
$$

After we had the unitary transformation of a generator with two indices, we could generalize it for a generator with only one index. For example, momentum operator.

$$
U(\Lambda)^{-1}P^\mu U(\Lambda) = \Lambda^\mu{}_\nu P^\nu \ .
$$

We can also see the commutation relation of momentum and other operators.

$$
\begin{aligned}
U(\Lambda)^{-1}P^\mu U(\Lambda) &= \left(\mathbb{1} - \frac{i}{2\hbar}\delta\omega_{\alpha\beta}M^{\alpha\beta}\right)P^\mu\left(\mathbb{1} + \frac{i}{2\hbar}\delta\omega_{\rho\sigma}M^{\rho\sigma}\right) \\
&= P^\mu + \frac{i}{2\hbar}\delta\omega_{\rho\sigma}\big[P^\mu, M^{\rho\sigma}\big] + \dots \ , \\
\Lambda^\mu{}_\nu P^\nu &= \big(\delta^\mu{}_\nu + \delta\omega^\mu{}_\nu\big)P^\nu \\
&= P^\mu + g^{\mu\rho}\delta\omega_{\rho\sigma}P^\sigma \ ,
\end{aligned}
$$

then,

$$
\begin{aligned}
\delta\omega_{\rho\sigma}\frac{i}{2\hbar}\big[P^\mu, M^{\rho\sigma}\big] &= \delta\omega_{\rho\sigma}g^{\mu\rho}P^\sigma \\
&= \frac{1}{2}\big(\delta\omega_{\rho\sigma}g^{\mu\rho}P^\sigma + \delta\omega_{\rho\sigma}g^{\mu\rho}P^\sigma\big) \\
&= \frac{1}{2}\delta\omega_{\rho\sigma}\big(g^{\mu\rho}P^\sigma - g^{\mu\sigma}P^\rho\big)
\end{aligned}
$$

by equating the antisymmetric coefficients,

$$
\big[P^\mu, M^{\rho\sigma}\big] = i\hbar\big(g^{\mu\rho}P^\sigma - g^{\mu\sigma}P^\rho\big) \ .
$$

Let us multiply with the angular momentum factor.

$$\begin{aligned}\big[J_i, P_j\big] &= \frac{1}{2}\varepsilon_{iab}\big[M^{ab}, P_j\big] \\ &= \frac{i\hbar}{2}\varepsilon_{iab}\big(\delta^j{}_b P_a - \delta^j{}_a P_b\big) \\ &= \frac{i\hbar}{2}\big(\varepsilon_{iaj}P_a - \varepsilon_{ijb}P_b\big) \\ &= -i\hbar\varepsilon_{ijk}P_k \ .\end{aligned}$$

We can also see the commutation relation with hamiltonian H or in another words, the zeroth component of momentum.

$$[J_i, P^0] = \frac{i\hbar}{2}\varepsilon_{iab}\big(g^{0b}P^a - g^{0a}P^b\big) = 0 \ .$$

Now, let's see the commutation relation between boost operator, momentum operator, and hamiltonian operator.

$$\big[K_i, P_j\big] = i\hbar\big(g^{j0}P^i - g^{ji}P^0\big) = -i\hbar\delta_{ij}H \ .$$

$$[K_i, H] = i\hbar\big(g^{00}P^i - g^{0i}P^0\big) = -i\hbar P_i \ .$$

The components of P should be able to commute with each other

$$\big[P^\mu, P^\nu\big] = 0 \ .$$

All of the commutation relation of operator that we have derived form the Lie algebra for the **Poincare group**.

Now let us consider the Lorentz transformation of a scalar quantum field. We can take the analogy straight from time dependent wave equation seen from the Heisenberg picture with futher relativistic generalization.

$$\exp\left(\frac{-iPx}{\hbar}\right)\varphi(0)\exp\left(\frac{iPx}{\hbar}\right) = \varphi(x) \ .$$

Define unitary spacetime translation operator as

$$T(a) \equiv \exp\left(\frac{-iP^\mu x^\mu}{\hbar}\right) \ .$$

Then we have

$$T(a)^{-1}\varphi(x)T(a) = \varphi(x-a) \ .$$

And for the infinitesimal translation

$$T(\delta a) = I - \frac{i}{\hbar}\delta a_\mu P^\mu \ .$$

Comparing the infinitesimal translation and the previous infinitesimal Lorentz transformation, with $\bar{x} = \Lambda^{-1}x$, Lorentz transformation of a scalar field equation will be

$$U(\Lambda)^{-1}\varphi(x)U(\Lambda) = \varphi(\Lambda^{-1}x) \ .$$

What would happen if we apply derivative operator to the transformation.

$$\begin{aligned}\partial^\mu\big[U(\Lambda)^{-1}\varphi(x)U(\Lambda)\big] &= U(\Lambda)^{-1}\partial^\mu\varphi(x)U(\Lambda) \\ &= \partial^\mu\varphi(\Lambda^{-1}x) \ .\end{aligned}$$

Let us examine the derivative operator first. If we have the transformation $\bar{x} = \Lambda^{-1}x$ then, like we have seen in the first chapter that the derivatives follow.

$$\begin{aligned}\bar{\partial}^\mu &= (\Lambda^{-1})^\mu{}_\nu\partial^\nu \\ \partial^\mu &= \Lambda^\mu{}_\nu\bar{\partial}^\nu \ .\end{aligned}$$

So, no we have

$$U(\Lambda)^{-1}\partial^\mu\varphi(x)U(\Lambda) = \Lambda^\mu{}_\nu\bar{\partial}^\nu\varphi(\Lambda^{-1}x) \ .$$

If we contract the derivative operator with another derivative operator, we will get

$$U(\Lambda)^{-1}\partial_\mu\partial^\mu\varphi(x)U(\Lambda) = \partial_\mu\Lambda^\mu{}_\nu\bar{\partial}^\nu\varphi(\Lambda^{-1}x) \ .$$

We first can lower the indices of the derivative transformation.

$$\begin{aligned}g_{\rho\mu}\partial^\mu &= g_{\rho\mu}\Lambda^\mu{}_\nu\bar{\partial}^\nu \\ \partial_\rho &= \Lambda_{\rho\nu}g^{\nu\sigma}\bar{\partial}_\sigma \\ &= \Lambda_\rho{}^\sigma\bar{\partial}_\sigma \\ &= (\Lambda^{-1})^\sigma{}_\rho\bar{\partial}_\sigma\end{aligned}$$

so,

$$\begin{aligned}U(\Lambda)^{-1}\partial_\mu\partial^\mu\varphi(x)U(\Lambda) &= (\Lambda^{-1})^\sigma{}_\mu\bar{\partial}_\sigma\Lambda^\mu{}_\nu\bar{\partial}^\nu\varphi(\Lambda^{-1}x) \\ &= \delta^\sigma{}_\nu\bar{\partial}_\sigma\bar{\partial}^\nu\varphi(\Lambda^{-1}x) \\ &= \bar{\partial}_\mu\bar{\partial}^\mu\varphi(\Lambda^{-1}x) \ .\end{aligned}$$

Now, we can see that the Klein-Gordon equation is invariant under Lorentz transformation.

$$U(\Lambda)^{-1}\left(-\partial_\mu\partial^\mu + \frac{m^2}{\hbar^2c^2}\right)\varphi(x)U(\Lambda) = \left(-\bar{\partial}_\mu\bar{\partial}^\mu + \frac{m^2}{\hbar^2c^2}\right)\varphi(\Lambda^{-1}x) \ .$$

Now, we will see what would happen if we take infinitesimal Lorentz transformation for the scalar field.

$$
\begin{aligned}
U(\Lambda)^{-1}\varphi(x)U(\Lambda) &= \Big(\mathbb{1} - \frac{i}{2\hbar}\delta\omega_{\mu\nu}M^{\mu\nu}\Big)\varphi(x)\Big(\mathbb{1} + \frac{i}{2\hbar}\delta\omega_{\alpha\beta}M^{\alpha\beta}\Big) \\
&= \varphi(x) + \frac{i}{2\hbar}\delta\omega_{\mu\nu}\big[\varphi(x), M^{\mu\nu}\big] + ... \\
\\
\varphi\Big(\big(\Lambda^{-1}\big)^{\mu}{}_{\nu}x^{\nu}\Big) &= \varphi\big(\Lambda_{\nu}{}^{\mu}x^{\nu}\big) \\
&= \varphi(x) + \big(\Lambda_{\nu}{}^{\mu}x^{\nu} - x^{\mu}\big)\partial_{\mu}\varphi(x) \\
&= \varphi(x) + \delta\omega_{\nu}{}^{\mu}x^{\nu}\partial_{\mu}\varphi(x) \\
&= \varphi(x) + \delta\omega_{\nu\mu}x^{\nu}\partial^{\mu}\varphi(x)\ .
\end{aligned}
$$

Now, if we equate both sides,

$$
\begin{aligned}
\frac{i}{2\hbar}\delta\omega_{\mu\nu}\big[\varphi(x), M^{\mu\nu}\big] &= \delta\omega_{\nu\mu}x^{\nu}\partial^{\mu}\varphi(x) \\
&= \frac{1}{2}\big(\delta\omega_{\nu\mu}x^{\nu}\partial^{\mu}\varphi(x) + \delta\omega_{\nu\mu}x^{\nu}\partial^{\mu}\varphi(x)\big) \\
\frac{i}{\hbar}\delta\omega_{\mu\nu}\big[\varphi(x), M^{\mu\nu}\big] &= \delta\omega_{\mu\nu}x^{\mu}\partial^{\nu}\varphi(x) + \delta\omega_{\nu\mu}x^{\nu}\partial^{\mu}\varphi(x) \\
\big[\varphi(x), M^{\mu\nu}\big] &= \frac{\hbar}{i}\big[x^{\mu}\partial^{\nu} - x^{\nu}\partial^{\mu}\big]\varphi(x)
\end{aligned}
$$

if we then define $\mathcal{L}^{\mu\nu} = \frac{\hbar}{i}\big[x^{\mu}\partial^{\nu} - x^{\nu}\partial^{\mu}\big]$,

$$
\big[\varphi(x), M^{\mu\nu}\big] = \mathcal{L}^{\mu\nu}\varphi(x)\ .
$$

Let us examine another commutation relation

$$
\begin{aligned}
\big[\big[\varphi(x), M^{\mu\nu}\big], M^{\rho\sigma}\big] &= \big[\mathcal{L}^{\mu\nu}\varphi(x), M^{\rho\sigma}\big] \\
&= \mathcal{L}^{\mu\nu}\big[\varphi(x), M^{\rho\sigma}\big] + \big[\mathcal{L}^{\mu\nu}, M^{\rho\sigma}\big]\varphi(x) \\
&= \mathcal{L}^{\mu\nu}\mathcal{L}^{\rho\sigma}\varphi(x) + \big[\mathcal{L}^{\mu\nu}, M^{\rho\sigma}\big]\varphi(x) \\
&= \mathcal{L}^{\mu\nu}\mathcal{L}^{\rho\sigma}\varphi(x) + \mathcal{L}^{\mu\nu}M^{\rho\sigma}\varphi(x) - M^{\rho\sigma}\mathcal{L}^{\mu\nu}\varphi(x) \\
&= \mathcal{L}^{\mu\nu}\mathcal{L}^{\rho\sigma}\varphi(x) + M^{\rho\sigma}\mathcal{L}^{\mu\nu}\varphi(x) - M^{\rho\sigma}\mathcal{L}^{\mu\nu}\varphi(x) \\
&= \mathcal{L}^{\mu\nu}\mathcal{L}^{\rho\sigma}\varphi(x)\ .
\end{aligned}
$$

Using the Jacobi identity, that is, $[[A,B],C] + [[B,C],A] + [[C,A],B] = 0$, we could examine

$$
\begin{aligned}
\big[\varphi(x), \big[M^{\mu\nu}, M^{\rho\sigma}\big]\big] &= \big[\big[M^{\rho\sigma}, \psi(x)\big], M^{\mu\nu}\big] + \big[\big[\psi(x), M^{\mu\nu}\big], M^{\rho\sigma}\big] \\
&= \big[\mathcal{L}^{\mu\nu}\varphi(x), M^{\rho\sigma}\big] - \big[\mathcal{L}^{\rho\sigma}\varphi(x), M^{\mu\nu}\big] \\
&= \mathcal{L}^{\mu\nu}\mathcal{L}^{\rho\sigma}\varphi(x) - \mathcal{L}^{\rho\sigma}\mathcal{L}^{\mu\nu}\varphi(x) \\
&= \big(\mathcal{L}^{\mu\nu}\mathcal{L}^{\rho\sigma} - \mathcal{L}^{\rho\sigma}\mathcal{L}^{\mu\nu}\big)\varphi(x)\ .
\end{aligned}
$$

Before we start to expand the right hand side, let it be known that

$$\partial^\alpha\big(x^\beta\partial^\gamma\big) = \big(\partial^\alpha x^\beta\big)\partial^\gamma + x^\beta\partial^\alpha\partial^\beta$$
$$= g^{\alpha\beta}\partial^\gamma + x^\beta\partial^\alpha\partial^\gamma \ .$$

$$\begin{aligned}\mathcal{L}^{\mu\nu}\mathcal{L}^{\rho\sigma}\varphi(x) &= -\hbar^2\big(x^\mu\partial^\nu - x^\nu\partial^\mu\big)\big(x^\rho\partial^\sigma - x^\sigma\partial^\rho\big)\psi(x)\\ &= -\hbar^2\big(x^\mu\partial^\nu x^\rho\partial^\sigma - x^\mu\partial^\nu x^\sigma\partial^\rho - x^\nu\partial^\mu x^\rho\partial^\sigma + x^\nu\partial^\mu x^\sigma\partial^\rho\big)\varphi(x)\\ &= -\hbar^2\big(g^{\nu\rho}x^\mu\partial^\sigma + x^\mu x^\rho\partial^\nu\partial^\sigma - g^{\nu\sigma}x^\mu\partial^\rho - x^\mu x^\sigma\partial^\nu\partial^\rho\\ &\quad -g^{\mu\rho}x^\nu\partial^\sigma - x^\nu x^\rho\partial^\mu\partial^\sigma + g^{\mu\sigma}x^\nu\partial^\rho + x^\nu x^\sigma\partial^\mu\partial^\rho\big)\varphi(x) \ .\end{aligned}$$

$$\begin{aligned}-\mathcal{L}^{\rho\sigma}\mathcal{L}^{\mu\nu}\varphi(x) &= \hbar^2\big(g^{\sigma\mu}x^\rho\partial^\nu + x^\rho x^\mu\partial^\sigma\partial^\nu - g^{\sigma\nu}x^\rho\partial^\mu - x^\rho x^\nu\partial^\sigma\partial^\mu\\ &\quad -g^{\rho\mu}x^\sigma\partial^\nu - x^\sigma x^\mu\partial^\rho\partial^\nu + g^{\rho\nu}x^\sigma\partial^\mu + x^\sigma x^\nu\partial^\rho\partial^\mu\big)\varphi(x) \ .\end{aligned}$$

So,

$$\mathcal{L}^{\mu\nu}\mathcal{L}^{\rho\sigma} - \mathcal{L}^{\rho\sigma}\mathcal{L}^{\mu\nu}$$
$$= i\hbar\big(g^{\mu\rho}\mathcal{L}^{\nu\sigma} - g^{\nu\rho}\mathcal{L}^{\mu\sigma} - g^{\mu\sigma}\mathcal{L}^{\nu\rho} + g^{\nu\sigma}\mathcal{L}^{\mu\rho}\big) \ .$$

Then,

$$\begin{aligned}\big[\varphi(x),\big[M^{\mu\nu},M^{\rho\sigma}\big]\big] &= \big(\mathcal{L}^{\mu\nu}\mathcal{L}^{\rho\sigma} - \mathcal{L}^{\rho\sigma}\mathcal{L}^{\mu\nu}\big)\varphi(x)\\ &= i\hbar\big(g^{\mu\rho}\mathcal{L}^{\nu\sigma} - g^{\nu\rho}\mathcal{L}^{\mu\sigma} - g^{\mu\sigma}\mathcal{L}^{\nu\rho} + g^{\nu\sigma}\mathcal{L}^{\mu\rho}\big)\varphi(x)\\ &= i\hbar\big(g^{\mu\rho}[\psi(x),M^{\nu\sigma}] - g^{\nu\rho}\big[\psi(x),M^{\mu\sigma}\big] - g^{\mu\sigma}\big[\psi(x),M^{\nu\rho}\big] + g^{\nu\sigma}\big[\psi(x),M^{\mu\rho}\big]\big)\\ &= \big[\varphi(x),i\hbar\big(g^{\mu\rho}M^{\nu\sigma} - g^{\nu\rho}M^{\mu\sigma} - g^{\mu\sigma}M^{\nu\rho} + g^{\nu\sigma}M^{\mu\rho}\big)\big] \ .\end{aligned}$$

Let us write

$$\Lambda^\rho{}_\tau = \delta^\rho{}_\tau + \frac{i}{2\hbar}\delta\omega_{\mu\nu}\Big(S_V{}^{\mu\nu}\Big)^\rho{}_\tau \ ,$$

where

$$\Big(S_V{}^{\mu\nu}\Big)^\rho{}_\tau \equiv \frac{\hbar}{i}\big(g^{\mu\rho}\delta^\nu{}_\tau - g^{\nu\rho}\delta^\mu{}_\tau\big)$$

are matrices which constitute the vector representation of the Lorentz generators. Then we examine the transformation of field function derivative using infinitesimal Lorentz transformation.

$$\begin{aligned}U(\Lambda)^{-1}\partial^\mu\varphi(x)U(\Lambda) &= \Big(\mathbb{1} - \frac{i}{2\hbar}\delta\omega_{\alpha\beta}M^{\alpha\beta}\Big)\partial^\mu\varphi(x)\Big(\mathbb{1} + \frac{i}{2\hbar}\delta\omega_{\rho\sigma}M^{\rho\sigma}\Big)\\ &= \partial^\mu\varphi(x) + \frac{i}{2\hbar}\delta\omega_{\alpha\beta}\big[\partial^\mu\varphi(x),M^{\alpha\beta}\big] + ...\end{aligned}$$

$$
\begin{aligned}
\partial^\mu \varphi(\Lambda^{-1}x) &= \partial^\mu \varphi\big(\Lambda_\sigma{}^\rho x^\sigma\big) \\
&= \partial^\mu \big(\varphi(x) + \big(\Lambda_\sigma{}^\rho x^\sigma - x^\rho\big)\partial_\rho \varphi(x)\big) \\
&= \partial^\mu \left(\varphi(x) + \frac{i}{2\hbar}\delta\omega_{\alpha\beta}\big(S^{\alpha\beta}\big)_\sigma{}^\rho x^\sigma \partial_\rho \varphi(x)\right) \\
&= \partial^\mu \varphi(x) + \frac{i}{2\hbar}\delta\omega_{\alpha\beta}\big(S^{\alpha\beta}\big)_\sigma{}^\rho \partial^\mu\big(x^\sigma \partial_\rho \varphi(x)\big) \\
&= \partial^\mu \varphi(x) + \frac{i}{2\hbar}\delta\omega_{\alpha\beta}\big(S^{\alpha\beta}\big)_\sigma{}^\rho \big(g^{\mu\sigma}\partial_\rho \psi(x) + x^\sigma \partial^\mu \partial_\rho \psi(x)\big) \\
&= \partial^\mu \varphi(x) + \frac{i}{2\hbar}\delta\omega_{\alpha\beta}\left(\big(S^{\alpha\beta}\big)^\mu{}_\rho \partial^\rho \psi(x) + \frac{\hbar}{i}\big[g^{\alpha\sigma}\delta^\beta{}_\rho - g^{\beta\sigma}\delta^\alpha{}_\rho\big] x_\sigma \partial^\rho \partial^\mu \psi(x)\right) \\
&= \partial^\mu \varphi(x) + \frac{i}{2\hbar}\delta\omega_{\alpha\beta}\left(\big(S^{\alpha\beta}\big)^\mu{}_\rho \partial^\rho \psi(x) + \frac{\hbar}{i}\big[x^\alpha \partial^\beta - x^\beta \partial^\alpha\big] \partial^\mu \psi(x)\right) \\
&= \partial^\mu \varphi(x) + \frac{i}{2\hbar}\delta\omega_{\alpha\beta}\left(\big(S^{\alpha\beta}\big)^\mu{}_\rho \partial^\rho \psi(x) + \mathcal{L}^{\alpha\beta}\partial^\mu \psi(x)\right) .
\end{aligned}
$$

So,

$$
\big[\partial^\mu \varphi(x), M^{\alpha\beta}\big] = \big(S^{\alpha\beta}\big)^\mu{}_\rho \partial^\rho \psi(x) + \mathcal{L}^{\alpha\beta}\partial^\mu \psi(x) .
$$

Let us examine how the matrix S commutes with each other.

$$
\begin{aligned}
\big[S^{\mu\nu}, S^{\alpha\beta}\big]^\rho{}_\sigma &= \big(S^{\mu\nu}\big)^\rho{}_\tau \big(S^{\alpha\beta}\big)^\tau{}_\sigma - \big(S^{\alpha\beta}\big)^\rho{}_\tau \big(S^{\mu\nu}\big)^\tau{}_\sigma \\
&= -\hbar^2\big(g^{\mu\rho}\delta^\nu{}_\tau - g^{\nu\rho}\delta^\mu{}_\tau\big)\big(g^{\alpha\tau}\delta^\beta{}_\sigma - g^{\beta\tau}\delta^\alpha{}_\sigma\big) + \hbar^2\big(g^{\alpha\rho}\delta^\beta{}_\tau - g^{\beta\rho}\delta^\alpha{}_\tau\big)\big(g^{\mu\tau}\delta^\nu{}_\sigma - g^{\nu\tau}\delta^\mu{}_\sigma\big) \\
&= -i\hbar\Big(g^{\mu\rho}\big(S^{\alpha\beta}\big)^\nu{}_\sigma - g^{\nu\rho}\big(S^{\alpha\beta}\big)^\mu{}_\sigma\Big) + i\hbar\Big(g^{\alpha\rho}\big(S^{\mu\nu}\big)^\beta{}_\sigma - g^{\beta\rho}\big(S^{\mu\nu}\big)^\alpha{}_\sigma\Big) \\
&= i\hbar\Big(g^{\alpha\rho}\big(S^{\mu\nu}\big)^\beta{}_\sigma - g^{\beta\rho}\big(S^{\mu\nu}\big)^\alpha{}_\sigma - g^{\mu\rho}\big(S^{\alpha\beta}\big)^\nu{}_\sigma + g^{\nu\rho}\big(S^{\alpha\beta}\big)^\mu{}_\sigma\Big)
\end{aligned}
$$

3. Canonical Quantization of Scalar Fields

Before we go back to the hamiltonian from section 1, let us set

$$c = \hbar = 1$$

for our convenience. Those factors could be reattained using dimensional analysis if needed. Now let us go back to previously constructed hamiltonian. For the case of free particles (no external potential working on the particle),

$$H = \int \mathrm{d}^3 x a^\dagger(\boldsymbol{x}) \left(-\frac{1}{2m} \nabla^2 \right) a(\boldsymbol{x}) \ .$$

Using inverse Fourier transformation on the integral, we can set the integral on momentum space,

$$a(\boldsymbol{x}) = \int \frac{\mathrm{d}^3 p'}{(2\pi)^{\frac{3}{2}}} \exp(-i\boldsymbol{p}' \cdot \boldsymbol{x}) \tilde{a}(\boldsymbol{p}') \ .$$

So, substituting back to the previous hamiltonian,

$$\begin{aligned}
H &= \int \mathrm{d}^3 x \, \mathrm{d}^3 p'' \, \mathrm{d}^3 p' \frac{1}{(2\pi)^3} \exp(i\boldsymbol{p}'' \cdot \boldsymbol{x}) \tilde{a}^\dagger(\boldsymbol{p}'') \left(-\frac{1}{2m} \nabla^2 \right) \exp(-i\boldsymbol{p}' \cdot \boldsymbol{x}) \tilde{a}(\boldsymbol{p}') \\
&= \int \mathrm{d}^3 x \, \mathrm{d}^3 p'' \, \mathrm{d}^3 p' \frac{1}{(2\pi)^3} \exp(i\boldsymbol{p}'' \cdot \boldsymbol{x}) \tilde{a}^\dagger(\boldsymbol{p}'') \left(-\frac{1}{2m} (-i\boldsymbol{p}')^2 \right) \exp(-i\boldsymbol{p}' \cdot \boldsymbol{x}) \tilde{a}(\boldsymbol{p}') \\
&= \int \mathrm{d}^3 p'' \, \mathrm{d}^3 p' \, \mathrm{d}^3 x \frac{1}{(2\pi)^3} \exp(i(\boldsymbol{p}'' - \boldsymbol{p}') \cdot \boldsymbol{x}) \frac{1}{2m} (\boldsymbol{p}')^2 \tilde{a}^\dagger(\boldsymbol{p}'') \exp(-i\boldsymbol{p}' \cdot \boldsymbol{x}) \tilde{a}(\boldsymbol{p}') \\
&= \int \mathrm{d}^3 p'' \, \mathrm{d}^3 p' \delta(\boldsymbol{p}'' - \boldsymbol{p}') \frac{1}{2m} (\boldsymbol{p}')^2 \tilde{a}^\dagger(\boldsymbol{p}'') \tilde{a}(\boldsymbol{p}') \\
&= \int \mathrm{d}^3 p \frac{1}{2m} \boldsymbol{p}^2 \tilde{a}^\dagger(\boldsymbol{p}) \tilde{a}(\boldsymbol{p}) \ .
\end{aligned}$$

The commutation (anticommutation) relations for the operators are

$$\begin{aligned}
\left[\tilde{a}(\boldsymbol{p}), \tilde{a}(\boldsymbol{p}') \right]_\mp &= 0 \\
\left[\tilde{a}^\dagger(\boldsymbol{p}), \tilde{a}^\dagger(\boldsymbol{p}') \right]_\mp &= 0 \\
\left[\tilde{a}(\boldsymbol{p}), \tilde{a}^\dagger(\boldsymbol{p}') \right]_\mp &= \delta(\boldsymbol{p} - \boldsymbol{p}')
\end{aligned}$$

Like the annhilation operator,

$$\tilde{a}(\boldsymbol{p}) |0\rangle = 0 \ .$$

We could see from the hamiltonian that energy is defined as $E(\boldsymbol{p}) = \frac{\boldsymbol{p}^2}{2m}$. If we want to generalize it to the relativistic limits, we could set the energy using the energy-momentum relation, that is $E(\boldsymbol{p}) = \sqrt{\boldsymbol{p}^2 c^2 + m^2 c^4}$. Substituting the energy back into the hamiltonian,

$$H = \int \mathrm{d}^3 p \sqrt{\boldsymbol{p}^2 c^2 + m^2 c^4} \tilde{a}^\dagger(\boldsymbol{p}) \tilde{a}(\boldsymbol{p}) \ .$$

This is the relativistic hamiltonian for free spin-0 particles (both bosons and fermions). To see if the theory is Lorentz invariant, let us go back to the Klein-Gordon equation, that is the Lorentz invariant equation of motion for classical scalar field.

$$(-\partial^2 + m^2)\varphi(x) = 0 \ .$$

Because φ is only classical field, the m is not yet to be mass as its dimension is $[L]^{-1}$. This equation of motion can be derived from the action principle $S = \int \mathrm{d}^4 x \mathcal{L}$, where $L = \int \mathrm{d}^3 x \mathcal{L}$ is the integral of lagrangian density $\mathcal{L}$. Action on another frame would be $\bar{S} = \int \mathrm{d}^4 x \bar{\mathcal{L}}$. Let us take a look at Lorentz transformation.

$$\begin{aligned}
\bar{x}^\mu &= \Lambda^\mu{}_\nu x^\nu \\
\partial_\alpha \bar{x}^\mu &= \Lambda^\mu{}_\nu \partial_\alpha x^\nu \\
\partial_\alpha \bar{x}^\mu &= \Lambda^\mu{}_\nu \delta_\alpha{}^\nu \\
\Lambda^\mu{}_\nu &= \partial_\nu \bar{x}^\mu \\
&= \frac{\partial \bar{x}^\mu}{\partial x^\nu} \ .
\end{aligned}$$

We can see that the Lorentz matrix also acts as the Jacobian between two inertial frame. So, the infinitesimal volume of the two frames are related with

$$\begin{aligned}
\mathrm{d}^4 \bar{x} &= \det(J)\, \mathrm{d}^4 x \\
&= \det(\Lambda)\, \mathrm{d}^4 x \\
&= \mathrm{d}^4 x \ .
\end{aligned}$$

Since we are taking consideration only of proper orthochronous transformation, infinitesimal volumes are the same between two frames. Now, if we want to make sure that action S is invariant, the only left is the lagrangian density. We then set that the lagrangian density must be the same in two frames

$$\bar{\mathcal{L}} = \mathcal{L} \ ,$$

so now,

$$\bar{S} = \int \mathrm{d}^4 \bar{x} \bar{\mathcal{L}} = \int \mathrm{d}^4 x \mathcal{L} = S \ .$$

Let us take the lagrangian density as

$$\mathcal{L} = -\frac{1}{2} \partial^\mu \varphi \partial_\mu \varphi - \frac{1}{2} m^2 \varphi^2 \ .$$

If we take the variation of action based on that lagrangian and apply the least action principle,

$$\delta S = 0$$

$$\int \mathrm{d}^4x\, \delta\mathcal{L} = 0$$

$$\int \mathrm{d}^4x \Big[-\frac{1}{2}\partial^\mu\, \delta\varphi \partial_\mu\varphi - \frac{1}{2}\partial^\mu\varphi\partial_\mu\, \delta\varphi - \frac{1}{2}m^2(2\varphi\, \delta\varphi) \Big] = 0$$

$$\int \mathrm{d}^4x \Big[-\frac{1}{2}\partial^\mu\, \delta\varphi \partial_\mu\varphi - \frac{1}{2}\partial^\mu\varphi\partial_\mu\, \delta\varphi - m^2\varphi\, \delta\varphi \Big] = 0$$

$$-\frac{1}{2}\Big[(\partial_\mu\varphi + \partial^\mu\varphi)(\delta\varphi)|_{-\infty}^{\infty} - 2\int \mathrm{d}^4x\, \delta\varphi\partial^\mu\partial_\mu\varphi \Big] - \int \mathrm{d}^4x m^2\varphi\, \delta\varphi = 0$$

$$\int \mathrm{d}^4x (\partial^\mu\partial_\mu - m^2)\varphi\, \delta\varphi = 0\ ,$$

so we obtained Klein-Gordon equation from the invariant lagrangian density. The general solution for Klein-Gordon is

$$\varphi(\boldsymbol{x}, t) = \int \frac{\mathrm{d}^3k}{f(k)} \big[a(\boldsymbol{k}) e^{i\boldsymbol{k}\cdot\boldsymbol{x} - i\omega t} + b(\boldsymbol{k}) e^{i\boldsymbol{k}\cdot\boldsymbol{x} + i\omega t} \big]\ .$$

We must impose that $\varphi(x) = \varphi^*(x)$ to ensure that the field equation is real.

$$\begin{aligned} \varphi^*(\boldsymbol{x}, t) &= \int \frac{\mathrm{d}^3k}{f(k)} \big[a^*(\boldsymbol{k}) e^{-i\boldsymbol{k}\cdot\boldsymbol{x} + i\omega t} + b^*(\boldsymbol{k}) e^{-i\boldsymbol{k}\cdot\boldsymbol{x} - i\omega t} \big] \\ &= \int \frac{\mathrm{d}^3k}{f(k)} \big[b^*(-\boldsymbol{k}) e^{i\boldsymbol{k}\cdot\boldsymbol{x} - i\omega t} + a^*(-\boldsymbol{k}) e^{i\boldsymbol{k}\cdot\boldsymbol{x} + i\omega t} \big]\ . \end{aligned}$$

Equating both fields, we get that $a(\boldsymbol{k}) = b^*(-\boldsymbol{k})$ and $b(\boldsymbol{k}) = a^*(-\boldsymbol{k})$. So, changing all coefficient to a, the field equation becomes

$$\begin{aligned} \varphi(\boldsymbol{x}, t) &= \int \frac{\mathrm{d}^3k}{f(k)} \big[a(\boldsymbol{k}) e^{i\boldsymbol{k}\cdot\boldsymbol{x} - i\omega t} + a^*(-\boldsymbol{k}) e^{i\boldsymbol{k}\cdot\boldsymbol{x} + i\omega t} \big] \\ &= \int \frac{\mathrm{d}^3k}{f(k)} \big[a(\boldsymbol{k}) e^{i\boldsymbol{k}\cdot\boldsymbol{x} - i\omega t} + a^*(\boldsymbol{k}) e^{-i\boldsymbol{k}\cdot\boldsymbol{x} + i\omega t} \big]\ . \end{aligned}$$

By defining 4-vector $k^\mu = (\omega, \boldsymbol{k})$,

$$\varphi(x) = \int \frac{\mathrm{d}^3k}{f(k)} \big[a(\boldsymbol{k}) e^{ik^\mu x_\mu} + a^*(\boldsymbol{k}) e^{-ik^\mu x_\mu} \big]\ .$$

We define **on-shell** momentum as $k^2 = -m^2$. So, in this case,

$$k^2 = -\omega^2 + \boldsymbol{k}^2 = -m^2\ ,$$

if we take $\exp i(\boldsymbol{k}\cdot\boldsymbol{x} \pm \omega t)$ as one solution of the Klein-Gordon equation.

Now we move on to choose the factor of $\frac{1}{f(k)}$ inside the integrand. An integration measure that is Lorentz (proper-orthochronous) invariant is $\int \mathrm{d}k^0\, \delta(k^2 + m^2)\theta(k^0)$ where δ and θ are the delta Dirac and step function.

$$\begin{aligned}\int_{-\infty}^{\infty} \mathrm{d}k^0\, \delta(k^2+m^2)\theta(k^0) &= \int_{-\infty}^{\infty} \mathrm{d}k^0\, \delta\Big(\boldsymbol{k}^2+m^2-\left(k^0\right)^2\Big)\theta(k^0) \\ &= \int_{-\infty}^{\infty} \mathrm{d}k^0\, \delta\Big(\omega^2-\left(k^0\right)^2\Big)\theta(k^0) \\ &= \int_{-\infty}^{\infty} \mathrm{d}k^0\, \delta(g(k^0))\theta(k^0) \\ &= \sum_i \frac{\theta({k^0}_i)}{|g'({k^0}_i)|} \\ &= \frac{1}{2\omega}\ .\end{aligned}$$

Then we can take

$$\frac{1}{f(k)} \equiv \frac{1}{(2\pi)^3 2\omega}$$

and also introduce a new momentum differential

$$\widetilde{dk} \equiv \frac{\mathrm{d}^3k}{(2\pi)^3 2\omega}\ .$$

Substituting both new quantities into the field equation, we would get the form

$$\begin{aligned}\varphi(x) &= \int \widetilde{\mathrm{d}k}\left[a(\boldsymbol{k})e^{ik^\mu x_\mu} + a^*(\boldsymbol{k})e^{-ik^\mu x_\mu}\right] \\ \int \mathrm{d}^3x e^{-ik'x}\varphi(x) &= \int \widetilde{\mathrm{d}k}\,\mathrm{d}^3x a(\boldsymbol{k})e^{i(k-k')x} + \int \widetilde{\mathrm{d}k}\,\mathrm{d}^3x a^*(\boldsymbol{k})e^{-i(k+k')x} \\ &= \int \widetilde{\mathrm{d}k}\,\mathrm{d}^3x e^{i(\boldsymbol{k}-\boldsymbol{k}')\cdot\boldsymbol{x}}e^{i(\omega'-\omega)t}a(\boldsymbol{k}) + \int \widetilde{\mathrm{d}k}\,\mathrm{d}^3x e^{-i(\boldsymbol{k}+\boldsymbol{k}')\cdot\boldsymbol{x}}e^{i(\omega'+\omega)t}a^*(\boldsymbol{k}) \\ &= \int \frac{\mathrm{d}^3k}{2\omega}\,\delta(\boldsymbol{k}-\boldsymbol{k}')e^{i(\omega'-\omega)t}a(\boldsymbol{k}) + \int \frac{\mathrm{d}^3k}{2\omega}\,\delta(\boldsymbol{k}'+\boldsymbol{k})e^{i(\omega'+\omega)t}a^*(\boldsymbol{k}) \\ &= \frac{1}{2\omega'}a(\boldsymbol{k}') + \frac{\exp(2i\omega' t)}{2\omega'}a^*(-\boldsymbol{k}')\ .\end{aligned}$$

Because we have the on-shell condition, that is $\omega = +\sqrt{\boldsymbol{k}^2+m^2}$, the integration over 3D delta Dirac also forces $\omega' = \omega$. Changing the parameter again,

$$\int \mathrm{d}^3x e^{-ikx}\varphi(x) = \frac{1}{2\omega}a(\boldsymbol{k}) + \frac{\exp(2i\omega t)}{2\omega}a^*(-\boldsymbol{k})\ .$$

Let us take the 0-th derivative of field equation.

$$\begin{aligned}\partial_0\varphi(x) &= i\omega\int \widetilde{\mathrm{d}k}\left[a(\boldsymbol{k})e^{ik^\mu x_\mu} - a^*(\boldsymbol{k})e^{-ik^\mu x_\mu}\right] \\ \int \mathrm{d}^3x e^{-ikx}\partial_0\varphi(x) &= \frac{i}{2}a(\boldsymbol{k}) + \frac{i\exp(2i\omega t)}{2}a(-\boldsymbol{k})\ .\end{aligned}$$

Now, we can combine both equation in order to express the coefficient a.

$$a(\boldsymbol{k}) = \int \mathrm{d}^3x \exp(-ikx)[i\partial_0\varphi(x) + \omega\varphi(x)],$$
$$a^*(\boldsymbol{k}) = \int \mathrm{d}^3x \exp(ikx)[-i\partial_0\varphi(x) + \omega\varphi(x)].$$

We now can start generalizing the hamiltonian. Remember that the classical hamiltonian is

$$H = \sum_i p_i\dot{q}_i - L,$$
$$p_i = \frac{\partial L}{\partial \dot{q}},$$

where p is the conjugate momentum and q is the generalized coordinates. The proper generalization to field theory would be that the continuous q is replaced by $\varphi(x)$ since our now lagrangian density is explicitly dependent on φ and its derivative. Then we have hamiltonian density expressed as

$$\mathcal{H} = \Pi\dot{\varphi} - \mathcal{L},$$
$$\Pi = \frac{\partial \mathcal{L}}{\partial \dot{\varphi}}.$$

Let us find the hamiltonian from the previous lagrangian.

$$\begin{aligned}\Pi &= \frac{\partial}{\partial\dot{\varphi}}\left(-\frac{1}{2}\partial^\mu\varphi\partial_\mu\varphi - \frac{1}{2}m^2\varphi^2\right)\\ &= \frac{\partial}{\partial\dot{\varphi}}\left(\frac{1}{2}\dot{\varphi}^2\right)\\ &= \dot{\varphi}\end{aligned}$$

$$\begin{aligned}\mathcal{H} &= \dot{\varphi}^2 + \frac{1}{2}\partial^\mu\varphi\partial_\mu\varphi + \frac{1}{2}m^2\varphi^2 - \Omega_0\\ &= \dot{\varphi}^2 - \frac{1}{2}\dot{\varphi}^2 + \frac{1}{2}(\boldsymbol{\nabla}\varphi)^2 + \frac{1}{2}m^2\varphi^2 - \Omega_0\\ &= \frac{1}{2}\dot{\varphi}^2 + \frac{1}{2}(\boldsymbol{\nabla}\varphi)^2 + \frac{1}{2}m^2\varphi^2 - \Omega_0.\end{aligned}$$

Integrating over volume, we could get the hamiltonian in terms of $a(\boldsymbol{k})$.

$$\begin{aligned}H &= \int \mathrm{d}^3x\mathcal{H}\\ &= -\Omega_0 V + \frac{1}{2}\int \widetilde{\mathrm{d}k}\omega[a^*(\boldsymbol{k})a(\boldsymbol{k}) + a(\boldsymbol{k})a^*(\boldsymbol{k})]\ .\end{aligned}$$

We now can transform the theory from classical to quantum mechanical using **canonical quantization**.

$$
\begin{aligned}
[\varphi(\boldsymbol{x},t),\varphi(\boldsymbol{x}',t)] &= 0 \ , \\
[\Pi(\boldsymbol{x},t),\Pi(\boldsymbol{x}',t)] &= 0 \ , \\
[\varphi(\boldsymbol{x},t),\Pi(\boldsymbol{x}',t)] &= i\,\delta(\boldsymbol{x}-\boldsymbol{x}') \ .
\end{aligned}
$$

We now can find the commutation relation between the operators a. Rewritting operator $a(\boldsymbol{k})$ using conjugate momentum, we get

$$
\begin{aligned}
a(\boldsymbol{k}) &= \int \mathrm{d}^3x e^{i(\omega t-\boldsymbol{k}\cdot\boldsymbol{x})}[i\Pi(\boldsymbol{x},t)+\omega\varphi(\boldsymbol{x},t)] \\
a^\dagger(\boldsymbol{k}') &= \int \mathrm{d}^3x' e^{i(\boldsymbol{k}'\cdot\boldsymbol{x}'-\omega t)}[-i\Pi(\boldsymbol{x}',t)+\omega\varphi(\boldsymbol{x}',t)] \ .
\end{aligned}
$$

Let us examine the commutation relation between nonconjugate operators with different momentum.

$$
\begin{aligned}
[a(\boldsymbol{k}),a(\boldsymbol{k}')] &= \int \mathrm{d}^3x\,\mathrm{d}^3x' e^{i(2\omega t-\boldsymbol{k}\cdot\boldsymbol{x}-\boldsymbol{k}'\cdot\boldsymbol{x}')} i\omega(\Pi\varphi'+\varphi\Pi'-\Pi'\varphi-\varphi'\Pi) \\
&= \int \mathrm{d}^3x\,\mathrm{d}^3x' e^{i(2\omega t-\boldsymbol{k}\cdot\boldsymbol{x}-\boldsymbol{k}'\cdot\boldsymbol{x}')} i\omega([\Pi,\varphi']+[\varphi,\Pi']) \\
&= \int \mathrm{d}^3x\,\mathrm{d}^3x' e^{i(2\omega t-\boldsymbol{k}\cdot\boldsymbol{x}-\boldsymbol{k}'\cdot\boldsymbol{x}')} i\omega(-i\,\delta(\boldsymbol{x}'-\boldsymbol{x})+i\,\delta(\boldsymbol{x}-\boldsymbol{x}')) \\
&= 0 \ .
\end{aligned}
$$

Now, for the conjugate operators,

$$
\begin{aligned}
\left[a^\dagger(\boldsymbol{k}),a^\dagger(\boldsymbol{k}')\right] &= \int ...(-\Pi\varphi'-\varphi\Pi'+\Pi'\varphi+\varphi'\Pi) \\
&= 0 \ .
\end{aligned}
$$

Lastly, for the conjugate with nonconjugate operators

$$
\begin{aligned}
\left[a(\boldsymbol{k}),a^\dagger(\boldsymbol{k}')\right] &= \int \mathrm{d}^3x\,\mathrm{d}^3x' e^{i(\boldsymbol{k}'\cdot\boldsymbol{x}'-\boldsymbol{k}\cdot\boldsymbol{x})} i\omega(\Pi\varphi'-\varphi\Pi'+\Pi'\varphi-\varphi'\Pi) \\
&= \int \mathrm{d}^3x\,\mathrm{d}^3x' e^{i(\boldsymbol{k}'\cdot\boldsymbol{x}'-\boldsymbol{k}\cdot\boldsymbol{x})} i\omega([\Pi,\varphi']+[\Pi',\varphi]) \\
&= \int \mathrm{d}^3x\,\mathrm{d}^3x' e^{i(\boldsymbol{k}'\cdot\boldsymbol{x}'-\boldsymbol{k}\cdot\boldsymbol{x})} i\omega(-2i\,\delta(\boldsymbol{x}-\boldsymbol{x}')) \\
&= 2\omega \int \mathrm{d}^3x e^{i\boldsymbol{x}\cdot(\boldsymbol{k}'-\boldsymbol{k})} \\
&= 2\omega(2\pi)^3\,\delta(\boldsymbol{k}-\boldsymbol{k}') \ .
\end{aligned}
$$

Collecting them all, we get the commutation relations

$$
\begin{aligned}
[a(\boldsymbol{k}),a(\boldsymbol{k}')] &= 0, \\
\left[a^\dagger(\boldsymbol{k}),a^\dagger(\boldsymbol{k}')\right] &= 0, \\
\left[a(\boldsymbol{k}),a^\dagger(\boldsymbol{k}')\right] &= 2\omega(2\pi)^3\,\delta(\boldsymbol{k}-\boldsymbol{k}') \ .
\end{aligned}
$$

Looking back at the integrated hamiltonian before,

$$\begin{aligned}
H &= -\Omega_0 V + \frac{1}{2}\int \widetilde{\mathrm{d}k}\omega\{a^\dagger(\boldsymbol{k}), a(\boldsymbol{k})\} \\
&= -\Omega_0 V + \int \widetilde{\mathrm{d}k}\omega a^\dagger(\boldsymbol{k})a(\boldsymbol{k}) - \frac{1}{2}\int \widetilde{\mathrm{d}k}\omega\left[a^\dagger(\boldsymbol{k}), a(\boldsymbol{k})\right] \\
&= -\Omega_0 V + \int \widetilde{\mathrm{d}k}\omega a^\dagger(\boldsymbol{k})a(\boldsymbol{k}) + \frac{1}{2}\int \frac{\mathrm{d}^3k}{(2\pi)^3 2\omega}\omega(2\omega(2\pi)^3\,\delta(0)) \\
&= -\Omega_0 V + \int \widetilde{\mathrm{d}k}\omega a^\dagger(\boldsymbol{k})a(\boldsymbol{k}) + (2\pi)^3\,\delta(0)\frac{1}{2}\int \frac{\mathrm{d}^3k}{(2\pi)^3}\omega \\
&= -\Omega_0 V + \int \widetilde{\mathrm{d}k}\omega a^\dagger(\boldsymbol{k})a(\boldsymbol{k}) + \varepsilon_0 V \\
&= (\varepsilon_0 - \Omega_0)V + \int \widetilde{\mathrm{d}k}\omega a^\dagger(\boldsymbol{k})a(\boldsymbol{k}),
\end{aligned}$$

where ε_0 is the integral over k-volume and V is $(2\pi)^3\,\delta(0)$. Let us examine the integration of ε_0 if we set the upper limit to some value of Λ. Assuming that $\Lambda \gg m$,

$$\begin{aligned}
\frac{1}{2(2\pi)^3}\int \mathrm{d}^3k\omega &= \frac{1}{2(2\pi)^3}4\pi\int_0^\Lambda \omega r_k^2\,\mathrm{d}r_k \\
&= \frac{1}{4\pi^2}\int_0^\Lambda \omega(\omega^2 - m^2)\,\mathrm{d}(\omega^2 - m^2)^{\frac{1}{2}} \\
&= \frac{1}{4\pi^2}\int_0^\Lambda \omega^2\sqrt{\omega^2 - m^2}\,\mathrm{d}\omega \\
&= \frac{1}{4\pi^2}\frac{1}{8}\left[\Lambda(2\Lambda^2 - m^2)\sqrt{\Lambda^2 - m^2} - m^4\ln\left(\frac{\Lambda + \sqrt{\Lambda^2 - m^2}}{m}\right)\right] \\
&\approx \frac{\Lambda^4}{16\pi^2}\ .
\end{aligned}$$

The justification of this approximation is that the formalism of quantum field will break down at certain large energy level. The Λ value is called the **ultraviolet cutoff**. But for now, it is sufficient to treat both Ω_0 and ε_0 as some arbitrary values. So we can even choose $\Omega_0 = \varepsilon_0$ thus eliminating the first term in the hamiltonian. We now have **re-derived** the hamiltonian for free relativistic bosons. The parameter m now can be seen as particle's mass instead of an arbitrary value.

Now, let us take a state of the form

$$|k_1 ... k_n\rangle \equiv a^\dagger(\boldsymbol{k}_1)...a^\dagger(\boldsymbol{k}_n)|0\rangle$$

and the $|0\rangle$ vacuum is annihilated by $a(\boldsymbol{k})|0\rangle$. We can apply this to the quantum field hamiltonian that we have derived earlier.

$$
\begin{aligned}
H|k_1...k_n\rangle &= \int \widetilde{\mathrm{d}k}\omega a^\dagger(\boldsymbol{k})a(\boldsymbol{k})a^\dagger(\boldsymbol{k}_1)...a^\dagger(\boldsymbol{k}_n)|0\rangle \\
&= \int \widetilde{\mathrm{d}k}\omega a^\dagger(\boldsymbol{k})\big[a(\boldsymbol{k}),a^\dagger(\boldsymbol{k}_1)\big]...a^\dagger(\boldsymbol{k}_n)|0\rangle + \int \widetilde{\mathrm{d}k}\omega a^\dagger(\boldsymbol{k})a^\dagger(\boldsymbol{k}_1)a(\boldsymbol{k})a^\dagger(\boldsymbol{k}_2)...a^\dagger(\boldsymbol{k}_n)|0\rangle \\
&= \int \mathrm{d}^3k\omega\,\delta(\boldsymbol{k}-\boldsymbol{k}_1)a^\dagger(\boldsymbol{k})a^\dagger(\boldsymbol{k}_2)...a^\dagger(\boldsymbol{k}_n)|0\rangle + \int \widetilde{\mathrm{d}k}\omega a^\dagger(\boldsymbol{k})a^\dagger(\boldsymbol{k}_1)a(\boldsymbol{k})a^\dagger(\boldsymbol{k}_2)...a^\dagger(\boldsymbol{k}_n)|0\rangle \\
&= \omega_1 a^\dagger(\boldsymbol{k}_1)...a^\dagger(\boldsymbol{k}_n)|0\rangle + ... + \omega_n a^\dagger(\boldsymbol{k}_1)...a^\dagger(\boldsymbol{k}_n)|0\rangle \\
&= (\omega_1 + ... + \omega_n)a^\dagger(\boldsymbol{k}_1)...a^\dagger(\boldsymbol{k}_n)|0\rangle \\
H|k_1...k_n\rangle &= \left[\sum_{i=1}^{n}\omega_i\right]|k_1...k_n\rangle\ .
\end{aligned}
$$

Thus, the state ket is an eigenstate of the hamiltonian.

Now, let us examine the Lorentz transformation of field equation again.

$$U(\Lambda)^{-1}\varphi(x)U(\Lambda) = \varphi(\Lambda^{-1}x)$$

$$\int \widetilde{\mathrm{d}k}\big[U(\Lambda)^{-1}a(\boldsymbol{k})U(\Lambda)e^{ikx} + U(\Lambda)^{-1}a^\dagger(\boldsymbol{k})U(\Lambda)e^{-ikx}\big] = \int \widetilde{\mathrm{d}k}\Big[a(\boldsymbol{k})e^{ik\Lambda^{-1}x} + a^\dagger(\boldsymbol{k})e^{-ik\Lambda^{-1}x}\Big],$$

let us examine the exponential part where k is contracted with a transformed x.

$$
\begin{aligned}
k^\mu(\Lambda^{-1}x)_\mu &= k^\mu(\Lambda^{-1})^\nu{}_\mu x_\nu \\
&= \Lambda_\mu{}^\nu k^\mu x_\nu \\
&= \Lambda^\mu{}_\nu k_\mu x^\nu \\
&= (\Lambda k)_\nu x^\nu\ .
\end{aligned}
$$

Now, we set $k \to \Lambda^{-1}k'$.

$$\int \widetilde{\mathrm{d}k}\big[U(\Lambda)^{-1}a(\boldsymbol{k})U(\Lambda)e^{ikx} + U(\Lambda)^{-1}a^\dagger(\boldsymbol{k})U(\Lambda)e^{-ikx}\big] = \int \widetilde{\mathrm{d}k'}\Big[a(\Lambda^{-1}\boldsymbol{k'})e^{ik'x} + a^\dagger(\Lambda^{-1}\boldsymbol{k'})e^{-ik'x}\Big].$$

Relabeling parameter k' to k, we now have

$$
\begin{aligned}
U(\Lambda)^{-1}a(\boldsymbol{k})U(\Lambda) &= a(\Lambda^{-1}\boldsymbol{k}) \\
U(\Lambda)^{-1}a^\dagger(\boldsymbol{k})U(\Lambda) &= a^\dagger(\Lambda^{-1}\boldsymbol{k}).
\end{aligned}
$$

Now, if we apply the unitary operator to the previous eigenstate,

$$U(\Lambda)|k_1...k_n\rangle = U(\Lambda)a^\dagger(\boldsymbol{k}_1)...a^\dagger(\boldsymbol{k}_n)|0\rangle,$$

Let us take a look at the transformation of the creation operator.

$$
\begin{aligned}
U(\Lambda)^{-1}a^\dagger(\boldsymbol{k})U(\Lambda) &= a^\dagger(\Lambda^{-1}\boldsymbol{k}) \\
U(\Lambda)^{-1}a^\dagger(\Lambda\boldsymbol{q})U(\Lambda) &= a^\dagger(\boldsymbol{q}) \\
a^\dagger(\Lambda\boldsymbol{q})U(\Lambda) &= U(\Lambda)a^\dagger(\boldsymbol{q}),
\end{aligned}
$$

so,

$$\begin{aligned} U(\Lambda)|k_1...k_n\rangle &= U(\Lambda)a^\dagger(\boldsymbol{k}_1)...a^\dagger(\boldsymbol{k}_n)|0\rangle \\ &= a^\dagger(\Lambda\boldsymbol{k}_1)...a^\dagger(\Lambda\boldsymbol{k}_n)U(\Lambda)|0\rangle \\ &= a^\dagger(\Lambda\boldsymbol{k}_1)...a^\dagger(\Lambda\boldsymbol{k}_n)|0\rangle \\ &= |\Lambda k_1...\Lambda k_n\rangle \ . \end{aligned}$$